GO SEARCH

AI-Powered Visual Product Discovery Platform

VIVA-VOCE PREPARATION GUIDE

Complete Question & Answer Reference for Project Defense

1. WHAT IS THIS PROJECT? (The Big Picture)

Go Search is a mobile application that lets you find products online by simply taking a photo or uploading an image — instead of typing words to search. Think of it like this: if you see a nice chair in a coffee shop and want to buy it, normally you'd struggle to describe it in words. With Go Search, you just take a photo and the app instantly finds similar products available for purchase.

The Core Idea in One Line:

"Show a photo, find the product" — no typing required.

Real-Life Scenario:

- You see someone wearing cool sunglasses.
- You don't know the brand name or style name.
- You open Go Search, take a photo, and the app finds those exact sunglasses (or very similar ones) for sale online.
- You can compare prices from different sellers and buy them — all in one app.

Who Is the Company Behind This Project?

This project is inspired by Pinterest, a company famous for its visual search technology called Pinterest Lens. Pinterest was founded in 2010 and has over 450 million users worldwide. They were one of the first to make visual search work at a massive scale. Go Search takes the same concept and builds it specifically for product shopping.

2. THE PROBLEM STATEMENT — Explained Simply

The problem this project solves is: Traditional keyword-based search fails when users cannot describe what they are looking for in text.

Why Is This a Problem?

Human beings recognize things visually first. When we see a product we like, our brain processes the image — shape, color, texture, style — before we can find words to describe it. But all existing search engines (Google, Amazon, Flipkart) require you to type words. This creates a huge gap:

- You see something visually → you like it → you want to buy it
- But you cannot describe it precisely in words
- So the search fails → you give up → the sale is lost

Who Suffers From This Problem?

Who	The Problem They Face
Customers	They see products in the real world (shops, Instagram, friend's house) but cannot find them online because they lack the vocabulary to describe them.
Small Sellers	Their unique, handmade, or niche products don't appear in keyword searches because they lack brand recognition and SEO expertise.
E-commerce Platforms	They lose sales when customers abandon searches due to inability to find products.

Common Viva Questions on Problem Statement

Q: What is the main problem your project solves?

A: The main problem is that traditional text-based search fails when users cannot describe what they want in words. People encounter products visually — in shops, social media, or real life — but struggle to translate that visual impression into a keyword search. Go Search eliminates this barrier by letting users search using images directly.

Q: Why can't users just use Google to search by image?

A: Google Images is a general-purpose search tool — it tells you what something IS, not where to BUY it. Go Search is built specifically for e-commerce. It not only identifies the product but also shows you where to buy it, compares prices across multiple sellers, and lets you complete the purchase — all in one app. Google does not integrate with seller inventories or shopping carts.

Q: Give me a real-life example of the problem.

A: Imagine you visit a friend's home and see a beautiful lamp on their table. You want to buy the same one. You don't know the brand, style name, or where they bought it. If you search 'modern lamp' on Amazon, you get thousands of results — none of which may be the exact one you saw. With Go Search, you photograph the lamp and the AI immediately finds it or very similar products for sale.

3. HOW THE AI TECHNOLOGY WORKS — Simple Explanation

The heart of Go Search is its AI-powered visual search pipeline. Here is exactly how it works, step by step:

Step-by-Step: What Happens When You Search With a Photo

Step	What Happens	Technology Used
Step 1	You take a photo or upload an image in the app.	React Native Camera / Gallery
Step 2	The image is sent to the AI server.	Node.js Backend API
Step 3	The AI model 'reads' the image and converts it into a list of numbers (called a feature vector or embedding) that mathematically represents the visual content — colors, shapes, textures, style.	TensorFlow / Vision Transformer (ViT)
Step 4	This list of numbers is compared against millions of product images already stored in the database (each product image was also converted to numbers when it was uploaded).	Pinecone Vector Database
Step 5	The products whose numbers are most similar to your query image are returned as results — ranked from most to least similar.	Approximate Nearest-Neighbor (ANN) Algorithm
Step 6	The results appear on your screen as a grid of product images with prices and seller info.	React Native UI

Key AI Concepts — Explained Simply

What is an Image Embedding / Feature Vector?

Imagine you describe a dog to someone who cannot see: 'It has 4 legs, is brown, medium-sized, has floppy ears.' You've converted a visual image into a description. An image embedding does the same thing — but using numbers instead of words. For example, an image might be converted to a list of 512 numbers like [0.23, 0.87, 0.11, ...]. These numbers capture the visual 'essence' of the image. Two images that look similar will produce similar number lists — that's how the AI finds matches.

What is a Vector Database (Pinecone)?

A normal database stores text and numbers and searches by exact match (e.g., 'find all products named Nike'). A vector database stores these image number-lists (embeddings) and can find the ones that are MOST SIMILAR — even if they're not identical. Pinecone is the specific vector database used in this project. It can search through millions of product embeddings in less than one second.

What is Approximate Nearest Neighbor (ANN) Search?

When you search for the most similar product, the computer needs to compare your image's numbers against ALL products in the database. If there are one million products, doing an exact comparison would take too long. ANN algorithms use smart shortcuts (like organizing products into groups) to find the closest matches very quickly — in under a second — without checking every single product. It trades a tiny bit of accuracy for a huge gain in speed.

What is a Vision Transformer (ViT)?

ViT is the specific AI model architecture used to convert images into embeddings. Unlike older models (like basic CNN/AlexNet) that looked at small patches of an image one at a time, ViT divides the image into patches and looks at ALL of them together simultaneously — understanding the relationship between different parts of the image. This makes it much better at understanding the overall style, context, and semantic meaning of an image — not just individual features.

Common Viva Questions on Technology

Q: How does the AI know which products are similar to my photo?

A: The AI converts both your query image and every product image into a 'feature vector' — a list of hundreds of numbers that mathematically represent the visual content of the image. Two visually similar images produce similar number lists. The system then finds the product vectors that are mathematically closest to your query vector using an algorithm called Approximate Nearest Neighbor search. The products with the closest vectors are the most visually similar matches.

Q: What is TensorFlow and why is it used?

A: TensorFlow is an open-source AI framework developed by Google. It provides the tools needed to build, train, and run deep learning models. In Go Search, TensorFlow is used to run the Vision Transformer model that converts images into embeddings. TensorFlow Serving specifically handles the deployment of this model as a fast, scalable service that the backend can call in real time.

Q: What is the difference between your approach and Amazon's image search?

A: Amazon's image search converts the query photo into text labels first (e.g., 'blue sneaker, athletic') and then runs those labels through its normal text search engine. This loses a lot of visual information — subtle style differences, exact color shades, or unique design elements that cannot be described in words. Go Search works entirely in the visual domain: it converts images to number vectors and compares those vectors directly, preserving all visual information and enabling more accurate style and appearance matching.

Q: What is Pinecone? Why not just use a normal database like MySQL?

A: Pinecone is a specialized vector database optimized for storing and searching high-dimensional number vectors (like image embeddings). A normal database like MySQL can find exact matches or sort by numbers, but it cannot efficiently find the 'most similar' vector among millions of options. Pinecone uses special data structures (like HNSW graphs) that organize vectors so that similar ones are stored near each other, enabling sub-second similarity search across millions of products — something MySQL simply cannot do at this scale.

4. SYSTEM ARCHITECTURE — The Technical Structure

The system is organized into 5 layers, each with a specific job:

Layer	What It Does	Technology
Presentation Layer	The mobile app interface that users see and interact with. Shows screens, buttons, images, and handles user input.	React Native
Application Layer	The 'brain' of the system. Receives requests from the app, coordinates the AI search, manages orders, users, and sellers.	Node.js Backend API
AI Layer	Converts images to embeddings and finds similar products. This is the core intelligence of the system.	TensorFlow + Pinecone
Data Layer	Stores all structured information: user profiles, product details, orders, seller info, reviews.	Supabase (PostgreSQL)
Infrastructure Layer	Cloud hosting, content delivery, monitoring — ensures the app works reliably for thousands of users simultaneously.	AWS / GCP + Docker

The Three User Roles

User Type	What They Can Do
Customer	Search by photo, browse products, compare prices, add to wishlist, buy products, track orders, use the AI chatbot.
Seller	List products (photos are auto-indexed by AI), manage inventory, view analytics on which products get the most search impressions.
Administrator	Monitor platform health, manage users, handle disputes, view system analytics and performance dashboards.

Common Viva Questions on Architecture

Q: Why did you choose React Native for the frontend?

A: React Native allows us to write a single codebase that runs on both iOS and Android. Without it, we would need to write the app twice — once in Swift for iOS and once in Kotlin for Android. React Native saves development time and ensures a consistent experience across both platforms. It also has strong community support and integrates well with our other chosen technologies.

Q: Why did you choose Supabase over Firebase?

A: Supabase provides a full PostgreSQL relational database, which is better for structured e-commerce data like orders, products, and user profiles — where relationships between data matter.

Firebase uses a NoSQL document model which is less suited for complex relational queries like 'show me all orders by this seller that are pending delivery.' Supabase also offers built-in authentication, storage, and real-time subscriptions, making it a complete backend-as-a-service solution.

Q: How does the seller's product become searchable?

A: When a seller uploads a product image, it is automatically sent through the same AI embedding pipeline as query images. The system generates a feature vector for the product image and stores it in the Pinecone vector database along with the product's ID and metadata (price, seller, etc.). This happens within 60 seconds of upload. After that, whenever a customer searches with a similar-looking image, the system will find and return this product in the results — without the seller needing to manually tag or categorize anything.

5. KEY FEATURES — Explained One by One

Feature 1: Visual Search (Core Feature)

The main feature. Users can search in three ways:

- Gallery Upload: Pick an existing photo from your phone.
- Camera Capture: Take a live photo of a product you see right now.
- URL Import: Paste a web link to a product image.

After submitting, results appear in under 2 seconds, ranked by visual similarity.

Feature 2: AI Chatbot Assistant

A built-in chat assistant that can help users find products by description, answer questions about products, help with order tracking, and provide personalized recommendations. It uses natural language processing to understand user messages and responds with relevant product suggestions or information.

Feature 3: Multi-Seller Price Comparison

When you search for a product visually, Go Search shows you the same or similar product from multiple different sellers simultaneously. You can see each seller's price, rating, and delivery options — allowing you to make an informed buying decision without switching between multiple websites.

Feature 4: Augmented Reality (AR) Preview

For furniture and home decor products, users can point their phone camera at a room and see a virtual preview of how the product would look in that space before buying. This uses the phone's camera and AR technology (ARKit on iPhone, ARCore on Android) to overlay a 3D model of the product onto the real-world view.

Feature 5: Personalized Discovery Feed

The home screen shows a personalized grid of products recommended specifically for each user, based on their past visual searches, wishlisted items, and purchase history. The more you use the app, the better the recommendations become, because the AI learns your visual style preferences.

Feature 6: Complete Shopping Flow

- Add to Cart and Checkout
- Multiple payment options: cards, UPI, cash on delivery
- Real-time order tracking with live map
- Wishlist with price change alerts
- Search history — re-run any past visual search with one tap

Common Viva Questions on Features

> **Q: What makes the AR preview feature useful in an e-commerce context?**

A: The biggest hesitation when buying furniture or home decor online is 'Will it fit in my room? Will it match my existing style?' AR preview lets customers place a virtual 3D model of the product in their actual room using their phone camera. They can see the size, color, and style in context before buying. This reduces purchase uncertainty, lowers return rates, and increases customer confidence — which directly improves conversion rates.

Q: How does the personalized feed work?

A: Every time a user submits a visual search, the system notes the visual characteristics of the query image and the results they clicked on. Over time, this builds a visual preference profile for each user — preferred colors, styles, price ranges, and product categories. The discovery feed uses this profile to surface products that match these learned preferences, even products the user hasn't explicitly searched for. The system improves continuously as more interaction data is collected.

6. EXISTING SYSTEM vs PROPOSED SYSTEM

Aspect	Existing Systems (Amazon, Flipkart)	Go Search (Proposed System)
Primary Search Method	Text keyword input required	Image-based input — no typing needed
How Visual Search Works	Image → converted to text labels → text search	Image → directly compared as visual vectors
Visual Information Preserved?	No — converted to text, richness is lost	Yes — full visual similarity is preserved
Small Seller Discoverability	Disadvantaged — needs SEO, brand recognition	Equal — any product discoverable by visual merit
Product Indexing	Manual tagging and categorization by seller	Automatic AI indexing on image upload
Personalization	Based on purchase/browsing history (text signals)	Based on visual preference signals
AR Preview	Not available in most platforms	Built-in for furniture and decor
AI Chatbot	Limited or scripted FAQ bots	Intelligent, context-aware shopping assistant

Disadvantages of Existing Systems

- Users who cannot describe a product in words cannot find it — leading to abandoned searches.
- Small sellers are invisible unless they invest in SEO optimization.
- Visual search is an add-on, not the core experience — it routes through text search and loses visual detail.
- No bridge between seeing a product in the physical world and finding it online.
- Product quality depends on how well the seller wrote the description — errors and omissions hide products.

Advantages of Go Search

- Completely eliminates the need to type — anyone can search, regardless of vocabulary.
- Preserves full visual information — finds products based on actual appearance, not text labels.
- Levels the playing field — independent sellers get discovered purely on visual merit.
- Automatic AI indexing — sellers just upload a good photo; the system does the rest.
- AR preview reduces purchase uncertainty and return rates.
- Scalable to millions of products while maintaining sub-second response times.

Common Viva Questions on Comparison

Q: Pinterest and Google Lens already do visual search. What is new about Go Search?

A: Pinterest Lens is primarily a content discovery tool for inspiration — it finds related pins and ideas, not specifically products to buy from multiple sellers with price comparison. Google Lens identifies objects but routes to general web search. Neither provides a complete end-to-end e-commerce experience: search, compare prices from multiple sellers, add to cart, checkout, and track delivery — all centered around visual search. Go Search is purpose-built as an e-commerce platform where visual search is the ONLY search paradigm, not an add-on feature.

Q: Why do you say small sellers benefit more from your system?

A: In keyword search, products rise to the top based on brand name recognition, advertising budget, review volume, and SEO optimization. A handmade artisan lamp with no brand name and a new seller account will never appear on the first page of Amazon search results, no matter how beautiful it is. In Go Search, that lamp is discovered purely because it looks similar to what a user photographed. Visual merit replaces brand authority as the ranking criterion, which is inherently fairer for small producers.

7. TECHNICAL SPECIFICATIONS & DESIGN

Hardware Requirements

Component	Specification	Why This is Needed
Processor	Intel Core i7 (13th Gen) or Apple M2	To handle AI model inference and running multiple development tools simultaneously.
RAM	16 GB minimum, 32 GB recommended	Running mobile emulators, development servers, and AI model testing at the same time needs a lot of memory.
GPU	NVIDIA RTX 3060 / Apple Metal	Accelerates TensorFlow model training and image embedding generation dramatically.
Storage	1 TB NVMe SSD	Stores training datasets, model weights, and all project code.
Mobile Devices	iPhone 13+ (iOS 16), Pixel 7 / Samsung S22 (Android 12)	For testing on real devices — emulators can't fully test camera, AR, and performance.

Software Stack

Software	Version	Purpose
React Native	0.73	Build the mobile app for both iOS and Android from one codebase.
TensorFlow / TF Lite	2.14	Run the image embedding AI model on the server and on mobile devices.
Node.js	20 LTS	Backend API — handles all business logic, coordinates between all services.
Supabase	Cloud Platform	PostgreSQL database, user authentication, and file storage.
Pinecone	Cloud Service	Vector database for storing and searching product image embeddings.
Figma	Latest	UI/UX design and wireframing for all 16 app screens.
Stripe	API	Secure payment processing for purchases.
Firebase	Cloud	Push notifications to users for order updates and price alerts.

The 16 Screens of the App

Group	Screen	Purpose
Onboarding	Splash Screen	Loading screen with animated logo. Auto-redirects returning users.
Onboarding	Onboarding Screen	3-step introduction for first-time users explaining the app's features.
Onboarding	Registration / Login	Sign up or log in via email, phone OTP, or Google/Apple.
Discovery	Home / Discovery Feed	Personalized product recommendations and quick access to visual search.
Discovery	Visual Search Interface	The main search screen — choose gallery, camera, or URL input.
Discovery	Camera Search Screen	Full-screen camera with live object detection and bounding box overlays.
Discovery	Image Results Page	Grid of visually similar products ranked by AI similarity score.
Product	Product Detail Page	Full product info — images, price, specs, reviews, seller info.
Product	Seller Storefront	Seller's profile — all their products, ratings, and follow option.
Commerce	Wishlist Screen	Saved products with price change tracking.
Commerce	Shopping Cart	Selected items, quantities, delivery fees, promo codes.
Commerce	Checkout Screen	Address, payment method, order review, and confirmation.
Account	Order Tracking	Live order status timeline and map tracking.
Account	User Profile	Account settings, saved addresses, payment methods.
Account	Notifications	Order updates, price alerts, new products from followed sellers.
Account	Search History / Settings	Visual log of past searches and app preferences.

8. TESTING & RESULTS

The project was tested across three major dimensions: visual search accuracy, performance under load, and security.

Visual Search Accuracy Testing

- An evaluation dataset of 5,000 product images was used across 5 categories: fashion, electronics, home decor, beauty, and sports.
- For each query image, a 'ground truth' set of correct matching products was manually identified.
- Three image quality levels were tested: professional photography, casual smartphone photos, and challenging conditions (low light, partial occlusion, unusual angles).
- Result — Top-5 Accuracy: 77.8% (the correct product appeared in the top 5 results 77.8% of the time).
- This exceeded the project target of 75% accuracy.

Performance Testing

- Tool Used: k6 load testing framework.
- Test: 500 concurrent users sending visual search queries simultaneously.
- Result: Response time under 2 seconds at the 95th percentile (P95) — meeting the project's target.
- Stress Test: System was gradually scaled to 5,000 concurrent users to identify the breaking point and verify graceful degradation (slows down before crashing, giving time to scale up).

Security Testing

- Tested against OWASP Mobile Application Security Verification Standard (MASVS).
- Tested against OWASP API Security Top 10.
- Specific tests: authentication bypass, injection attacks, rate limiting, data exposure vulnerabilities.

Common Viva Questions on Testing

Q: What is Top-5 accuracy and why is it used instead of Top-1?

A: Top-1 accuracy means the correct product is the very first result. Top-5 means the correct product appears somewhere in the first 5 results. For visual product search, Top-5 is a more realistic metric because users don't expect the exact product to always be first — they are willing to scroll through a few results. Our system achieved 77.8% Top-5 accuracy, meaning in about 78 out of 100 searches, the correct product is within the first 5 results shown. This is commercially viable for a real-world application.

Q: What does P95 response time mean?

A: P95 means the 95th percentile response time — the time within which 95% of all requests are completed. If the P95 is 2 seconds, it means 95 out of every 100 requests finish in under 2 seconds. The remaining 5% may take longer, but those are outliers. P95 is a standard industry metric for measuring application performance because it captures real-world user experience better than average response time, which can hide slow outliers.

Q: What is OWASP and why is it mentioned in testing?

A: OWASP (Open Web Application Security Project) is a globally recognized organization that publishes standard lists of the most critical security vulnerabilities in web and mobile applications. The OWASP Top 10 and MASVS are used as checklists by security testers worldwide. By testing against these standards, we ensure our application is protected against the most common and dangerous attack types — including SQL injection, authentication bypass, and data exposure — following industry-accepted best practices.

9. OBJECTIVES AND SCOPE

Project Objectives

User Experience Objectives

- Accept image inputs via gallery, camera, and URL.
- Return visually similar product results in under 2 seconds.
- Enable price comparison across multiple sellers without text input.
- Provide a complete purchase flow from visual search to order delivery.
- Be usable by anyone — regardless of tech skill level.

Technical Objectives

- Build an AI pipeline that extracts meaningful visual features from product images.
- Support a catalog of up to 1 million products with sub-second search.
- Index new seller products within 60 seconds of upload.
- Handle thousands of simultaneous users without performance degradation.

Business Objectives

- Help independent sellers reach visually-motivated buyers.
- Provide sellers with analytics on search impressions and conversions.
- Build an architecture that can expand to AR try-on, social sharing, and third-party logistics.

What Is IN Scope

- Mobile app for iOS and Android.
- Visual search with 3 input modalities.
- Multi-seller catalog with comparison view.
- Complete e-commerce flow: search → buy → track.
- Seller dashboard and admin panel.
- AI chatbot, AR preview, personalized feed.

What Is OUT of Scope (Future Work)

- Web browser version of the app.
- Physical point-of-sale system integration.
- Live streaming commerce.
- Cryptocurrency payments.
- Direct logistics and fulfillment management.

Common Viva Questions on Objectives & Scope

Q: Why is video search listed as future work and not implemented now?

A: Video-based visual search requires extracting representative frames from video, processing each frame through the embedding model, and aggregating the results — which is significantly more

computationally expensive than a single image search. For the initial version, we prioritized getting the core image search pipeline working accurately and efficiently. Video search is a valuable enhancement but adds considerable complexity that was scoped out to keep the initial development focused and achievable within the project timeline.

Q: How would the system scale if it grows to 10 million products?

A: The architecture is designed for this. Pinecone as the vector database scales horizontally — you simply add more pods to handle a larger index. The Node.js backend is containerized with Docker and deployed on cloud infrastructure, so new instances can be spun up automatically when load increases (auto-scaling). TensorFlow Serving for the embedding model can also be scaled to multiple GPU instances. The Supabase PostgreSQL database supports read replicas for handling high query volumes. Each layer of the system can be scaled independently based on demand.

10. QUICK-FIRE VIVA QUESTIONS — Short Answers

These are the most commonly asked one-liner type questions in a viva. Memorize these answers:

Question	Short Answer
What does 'embedding' mean?	A list of numbers that represents the visual content of an image — like a mathematical fingerprint.
What is a vector database?	A database that stores number-lists (vectors) and can find the most similar ones quickly.
What is ANN search?	Approximate Nearest Neighbor — a fast algorithm to find the most similar vectors without checking every single one.
Why 'approximate' and not 'exact'?	Exact search over millions of vectors is too slow. ANN is fast enough (under 1 second) with only a tiny loss in accuracy — an acceptable trade-off.
What is ViT?	Vision Transformer — the AI model architecture that converts images to embeddings by analyzing the relationships between all image patches simultaneously.
What is React Native?	A JavaScript framework that lets you build apps for both iOS and Android from a single codebase.
What is Supabase?	An open-source backend platform that provides a PostgreSQL database, authentication, and file storage.
What is Pinecone?	A cloud vector database service optimized for storing and searching high-dimensional embeddings.
What is cosine similarity?	A mathematical measure of similarity between two vectors — used to determine how visually similar two images are.
How many screens does the app have?	16 screens organized into 5 groups: Onboarding, Discovery, Product, Commerce, and Account.
What is the response time target?	Under 2 seconds for visual search results, even with 500 concurrent users.
What accuracy did the system achieve?	77.8% Top-5 accuracy, exceeding the 75% target.
What payment gateway is used?	Stripe — for secure credit/debit card transactions.
How are push notifications sent?	Via Firebase Cloud Messaging (FCM).
What is TensorFlow Lite?	A lightweight version of TensorFlow optimized for running AI models directly on mobile devices, reducing server load.
What categories does the app cover?	Fashion, Consumer Electronics, Home & Living, Beauty & Personal Care, Sports & Outdoor Equipment.
What is the indexing time for new products?	Within 60 seconds of a seller uploading a product image.

Question	Short Answer
What does 'zero-friction discovery' mean?	The core design philosophy: a user should go from seeing a product to finding it for sale in 3 interactions or fewer.
What is HNSW?	Hierarchical Navigable Small World — a graph-based algorithm used for fast approximate nearest-neighbor search in vector databases.
Who are the three types of users?	Customers (buyers), Sellers (merchants), and Administrators (platform operators).

11. FUTURE ENHANCEMENTS

These are the planned improvements for future versions of Go Search. Be ready to explain each one:

Enhancement	What It Means
Multi-Object Search	Currently, you search for one product at a time. Future: Point the camera at a table with a lamp, vase, and chair, and the app simultaneously searches for all three objects it detects in the frame.
Style & Aesthetic Transfer	Instead of finding the exact product, find products that match the overall 'vibe' or aesthetic of an image — same color palette, same design era, same mood. Useful for interior design and fashion styling.
3D Product Visualization	View 3D models of products — rotate, zoom, inspect from all angles — before purchasing. Combined with AR, this gives a near-physical shopping experience.
Fine-Grained Visual Similarity	Better at distinguishing subtle differences within a category — e.g., telling apart two very similar shoe styles or two similar fabric patterns that look nearly identical to the current model.
Video-Based Visual Search	Submit a short video clip (from Instagram, YouTube, etc.) as a search query. The system extracts frames, analyzes them, and finds products that appear in the video.

Q: If you had more time, what would you improve first?

A: I would prioritize Multi-Object Search because it has the highest practical impact for users. Right now, if you see a fully styled room you love, you have to search for each item separately. Multi-object search would let you photograph the entire room and instantly discover all visible products simultaneously — this would be a major usability leap and would significantly increase the app's utility for home decor and fashion discovery scenarios.

12. CONCLUSION — What the Project Achieved

This section is important for the end of your viva. Be ready to summarize what the project accomplished:

What Was Built:

- A fully functional 16-screen mobile application for AI-powered visual product discovery.
- A complete e-commerce workflow from visual search to order delivery tracking.
- An end-to-end AI pipeline: image input → embedding → vector search → ranked results.

What Was Achieved:

- 77.8% Top-5 accuracy — exceeded the 75% target.
- Sub-2-second response at P95 for 500 concurrent users — met the performance target.
- All 16 screens navigable with primary flows completable in under 3 minutes — met usability target.

Why It Matters:

- Bridges the gap between visual product recognition and online shopping.
- Democratizes product discoverability for independent sellers.
- Demonstrates practical application of state-of-the-art AI (Vision Transformers, vector databases) in a consumer mobile app.

Q: Summarize your project in 3 sentences for the examiner.

A: Go Search is an AI-powered mobile e-commerce app that enables users to find and purchase products by taking a photo, completely eliminating the need for keyword-based searching. The system uses a Vision Transformer model to convert images into numerical embeddings and a vector database (Pinecone) to find visually similar products in under 2 seconds. It achieved 77.8% Top-5 search accuracy in testing and provides a complete shopping experience including multi-seller price comparison, AR preview, personalized recommendations, and order tracking.

Best of Luck for Your Viva-Voce!

Remember: Speak confidently, use simple language, and relate every answer back to the core problem the project solves.